

Forty-cut continuation

Pine · 18 × 140 mm · usable lengths

plan-18f74717-20d0-48cc-9c2c-760fd83c4a81-13 · workspace revision 7 · plan created 9/3/2026, 10:34:47 AM (local browser time)

Saw kerf: 20 mm per part. **Keep positive remnants:** at least 500 mm.

Usable lengths only; deduct trimming and defects first. One saw pass per produced part, including the last one. Same material and cross-section throughout.

40 / 40 parts planned Optimal for this objective

Stock opened	Finished parts	Sawdust
100,000 mm	40,000 mm	800 mm
Reusable remnants	Short scrap	Boards handled
59,200 mm	0 mm	1

The cutting layout All lengths in mm

One-metre shelf Sawdust Keep Scrap

Lengths share one proportional scale across these boards. Thin kerf strips may be narrower than a screen pixel, and short segments may not fit a label; the exact cut order and remnant accounting give the full measurements.

stock-18f74717-20d0-48cc-9c2c-760fd83c4a81-9 **Long usable length** Full board 100,000 mm



40 parts · 800 mm sawdust 59,200 mm to keep

Order	Part	Length	Start	Kerf
1	One-metre shelf #1 part-18f74717-20d0-48cc-9c2c-760fd83c4a81-11	1,000 mm	0 mm	20 mm
2	One-metre shelf #2 part-18f74717-20d0-48cc-9c2c-760fd83c4a81-11	1,000 mm	1,020 mm	20 mm

Order	Part	Length	Start	Kerf
3	One-metre shelf #3 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	2,040 mm	20 mm
4	One-metre shelf #4 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	3,060 mm	20 mm
5	One-metre shelf #5 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	4,080 mm	20 mm
6	One-metre shelf #6 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	5,100 mm	20 mm
7	One-metre shelf #7 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	6,120 mm	20 mm
8	One-metre shelf #8 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	7,140 mm	20 mm
9	One-metre shelf #9 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	8,160 mm	20 mm
10	One-metre shelf #10 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	9,180 mm	20 mm
11	One-metre shelf #11 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	10,200 mm	20 mm
12	One-metre shelf #12 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	11,220 mm	20 mm

Order	Part	Length	Start	Kerf
13	One-metre shelf #13 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	12,240 mm	20 mm
14	One-metre shelf #14 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	13,260 mm	20 mm
15	One-metre shelf #15 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	14,280 mm	20 mm
16	One-metre shelf #16 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	15,300 mm	20 mm
17	One-metre shelf #17 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	16,320 mm	20 mm
18	One-metre shelf #18 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	17,340 mm	20 mm
19	One-metre shelf #19 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	18,360 mm	20 mm
20	One-metre shelf #20 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	19,380 mm	20 mm
21	One-metre shelf #21 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	20,400 mm	20 mm
22	One-metre shelf #22 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	21,420 mm	20 mm

Order	Part	Length	Start	Kerf
23	One-metre shelf #23 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	22,440 mm	20 mm
24	One-metre shelf #24 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	23,460 mm	20 mm
25	One-metre shelf #25 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	24,480 mm	20 mm
26	One-metre shelf #26 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	25,500 mm	20 mm
27	One-metre shelf #27 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	26,520 mm	20 mm
28	One-metre shelf #28 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	27,540 mm	20 mm
29	One-metre shelf #29 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	28,560 mm	20 mm
30	One-metre shelf #30 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	29,580 mm	20 mm
31	One-metre shelf #31 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	30,600 mm	20 mm
32	One-metre shelf #32 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	31,620 mm	20 mm

Order	Part	Length	Start	Kerf
33	One-metre shelf #33 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	32,640 mm	20 mm
34	One-metre shelf #34 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	33,660 mm	20 mm
35	One-metre shelf #35 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	34,680 mm	20 mm
36	One-metre shelf #36 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	35,700 mm	20 mm
37	One-metre shelf #37 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	36,720 mm	20 mm
38	One-metre shelf #38 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	37,740 mm	20 mm
39	One-metre shelf #39 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	38,760 mm	20 mm
40	One-metre shelf #40 part-18f74717- 20d0-48cc-9c2c- 760fd83c4a81-11	1,000 mm	39,780 mm	20 mm

Starts are measured from the usable stock end; each part includes one subsequent saw pass.

Every millimetre accounted for. 40.0% of opened stock becomes finished parts in this proposal.

$100,000 \text{ mm} = 40,000 + 800 + 59,200 + 0 \text{ mm}$

Waste is sawdust + short scrap (800 mm). Reusable remnants and untouched stock are not waste.

For a fixed complete job, sawdust is constant: one kerf per requested part. The trade-offs are stock opened, boards handled, and whether remaining lengths become reusable remnants or short scrap.

Every copy checked

40 planned / 40 requested

Part	Requested	Planned	Missing
One-metre shelf part -18f74717-20d0- 48cc-9c2c- 760fd83c4a81-11	40	40	0

Deterministic branch-and-bound. 1 of 100,000 search nodes examined. The search proved this solution optimal for the selected objective and allowed stock.

Minimize the total length of stock opened for this job. Ties favor less waste, then fewer boards.

A planning estimate, not machine-control instructions or a safe-cutting guarantee. Verify measurements, stock identity and safe workholding at the bench. Approval does not physically consume stock.